Customer Support Ticket Analyser

Project Overview

- Built a Python-based Ticket Analysis System to store, clean, and analyze customer support tickets.
- Implemented data cleaning, keyword analysis, and priority insights to identify common issues and improve service quality.

Project Objective

- To analyze customer support data and extract meaningful insights that help improve service quality, response time, and customer satisfaction.

Problem Statement

- Customer support teams handle numerous service tickets daily. Analysing support tickets helps identify common issues, customer sentiment, support quality, and areas for improvement.
- In this project, you will build a Python-based Ticket Analysis System that stores, cleans, analyses, and extracts insights from customer support tickets.

Dataset

The data set is organized in a structured format (dictionary/tabular form) with the following fields:

- Ticket_No: unique identifier assigned to each support ticket.
- Customer_Name: The name of the customer who raised the issue.
- Issue_Description: A textual description of the problem reported by the customer. This field contains unstructured text data, which is later cleaned and analyzed.
- Priority: Ticket priority categorized as: (High/Medium/Low)

Tools & Technologies Used

- Python - Core programming and logic implementation
- Built-in Libraries - String methods (.lower(), .replace(), .split(), .strip()) for text processing
- Google Colab

Data Cleaning Process

- Converted all issue descriptions to lowercase for consistency
- Replaced informal words (e.g., "ok" → "okay") for standardization
- Removed punctuation marks (., ! ? -) to clean the text
- Eliminated multiple spaces and converted them into a single space
- Trimmed leading and trailing spaces to improve readability
- Prepared clean and structured text data for accurate analysis

Code Implementation

- Created a dictionary-of-lists (ticket_data) to store ticket details.
- Added new tickets dynamically using loops, user input, and auto-increment logic.
- Cleaned issue descriptions using string methods (lower(), replace(), split(), strip()).
- Built a function for keyword-based search to analyze common issues.
- Performed priority counting and insights generation using loops and conditions.

Final Insights

- High-priority tickets (6) are highest, indicating more critical issues than others.
- Medium-priority tickets (5) are second highest, showing moderate issue volume.
- Low-priority tickets (4) are lowest, meaning fewer minor issues.
- Keywords like “slow” and “poor” highlight performance and service issues. "slow" → 2 tickets "poor" → 2 tickets
- Positive terms like “good” and “excellent” indicate some customer satisfaction. "good" → 2 tickets "excellent" → 1 ticket
- Ticket No: 2 (Meera) Issue: slow response, very poor service Longer descriptions indicate strong dissatisfaction. These tickets need priority investigation
- Majority of complaints are related to system performance and reliability
- Support team shows strong communication and resolution skills
- 53 unique words indicate diverse customer issues, not a single dominant problem.
- Overall customer experience is mixed with both complaints and appreciation

Recommendation

- Focus on reducing response time and resolving high-priority technical issues to significantly improve overall customer satisfaction.

Conclusion

- The Ticket Analysis System successfully identified key patterns in customer issues, highlighting that high-priority and performance-related problems (slow response, technical failures) are the main concerns.
- While customers appreciate support quality in some cases, improving response time and system reliability is essential to enhance overall customer satisfaction